

TRIGONOMETRY — I

Class 11 · Maharashtra Board · MHT-CET Foundation · Maths Chapter 2

MISMATHE Study Sheet · Mindmap + Formulas + Must-Remember · Test: 5 July

THE STANDARD-ANGLE TABLE

★★★ memorize this FIRST

θ	0°	30°	45°	60°	90°
$\sin \theta$	0	$1/2$	$1/\sqrt{2}$	$\sqrt{3}/2$	1
$\cos \theta$	1	$\sqrt{3}/2$	$1/\sqrt{2}$	$1/2$	0
$\tan \theta$	0	$1/\sqrt{3}$	1	$\sqrt{3}$	∞
$\operatorname{cosec} \theta$	∞	2	$\sqrt{2}$	$2/\sqrt{3}$	1
$\sec \theta$	1	$2/\sqrt{3}$	$\sqrt{2}$	2	∞
$\cot \theta$	∞	$\sqrt{3}$	1	$1/\sqrt{3}$	0

Trick: sin values = $\sqrt{0/2}, \sqrt{1/2}, \sqrt{2/2}, \sqrt{3/2}, \sqrt{4/2} \rightarrow 0, \frac{1}{2}, 1/\sqrt{2}, \sqrt{3}/2, 1$. cos = reverse. tan = sin/cos.

MINDMAP — The 6 Branches

1 · The 6 Functions & Relations

- Reciprocal:** $\operatorname{cosec} = 1/\sin$, $\sec = 1/\cos$, $\cot = 1/\tan$
- Quotient:** $\tan = \sin/\cos$, $\cot = \cos/\sin$
- Defined using a unit circle ($x=\cos\theta$, $y=\sin\theta$)

2 · Signs in Quadrants ASTC

- Q1 (All)** — all positive
- Q2 (Sin)** — sin & cosec +
- Q3 (Tan)** — tan & cot +
- Q4 (Cos)** — cos & sec +
- "Add Sugar To Coffee"

3 · Fundamental (Pythagorean) Identities ★ use everywhere

$$\sin^2\theta + \cos^2\theta = 1$$

$$1 + \tan^2\theta = \sec^2\theta$$

$$1 + \cot^2\theta = \operatorname{cosec}^2\theta$$

- Rearranged: $\sin^2\theta = 1 - \cos^2\theta$; $\sec^2\theta - \tan^2\theta = 1$; $\operatorname{cosec}^2\theta - \cot^2\theta = 1$

4 · Negative Angles (even/odd)

- Odd** (sign flips): $\sin(-\theta) = -\sin\theta$, $\tan(-\theta) = -\tan\theta$, cosec , $\cot$
- Even** (sign stays): $\cos(-\theta) = \cos\theta$, $\sec(-\theta) = \sec\theta$

5 · Periodicity

- sin, cos, cosec, sec $\rightarrow$ period 2π
- tan, cot $\rightarrow$ period π
- $\sin(\theta+2\pi) = \sin\theta$, $\tan(\theta+\pi) = \tan\theta$

6 · Allied Angles ★ most-asked concept

- RULE 1 (function):** for $90^\circ \pm \theta$ and $270^\circ \pm \theta \rightarrow$ function CHANGES ($\sin \leftrightarrow \cos$, $\tan \leftrightarrow \cot$, $\sec \leftrightarrow \operatorname{cosec}$). For $180^\circ \pm \theta$ and $360^\circ \pm \theta \rightarrow$ function STAYS SAME.
- RULE 2 (sign):** decide the sign by the quadrant the *whole* angle lands in (use ASTC).
- e.g. $\sin(180^\circ - \theta) = +\sin\theta$; $\cos(90^\circ + \theta) = -\sin\theta$; $\tan(270^\circ - \theta) = +\cot\theta$

DOMAIN & RANGE

sin	Domain: $\mathbb{R}$ Range: $[-1, 1]$	cos	Domain: $\mathbb{R}$ Range: $[-1, 1]$
tan	Domain: $\mathbb{R} - \{\text{odd} \times \pi/2\}$ Range: $\mathbb{R}$	cot	Domain: $\mathbb{R} - \{n\pi\}$ Range: $\mathbb{R}$
sec	Domain: $\mathbb{R} - \{\text{odd} \times \pi/2\}$ Range: $(-\infty, -1] \cup [1, \infty)$	csc	Domain: $\mathbb{R} - \{n\pi\}$ Range: $(-\infty, -1] \cup [1, \infty)$

MUST-REMEMBER

write these on one page

1	The standard-angle table (sin/cos/tan of 0,30,45,60,90) — cold
2	ASTC sign rule (All-Sin-Tan-Cos across Q1-Q2-Q3-Q4)
3	3 identities: $\sin^2+\cos^2=1$; $1+\tan^2=\sec^2$; $1+\cot^2=\operatorname{cosec}^2$
4	Reciprocals: $\operatorname{cosec}=1/\sin$, $\sec=1/\cos$, $\cot=1/\tan$; $\tan=\sin/\cos$
5	Negative angles: cos & sec are EVEN (stay), rest are ODD (flip sign)
6	Allied: $90^\circ/270^\circ \rightarrow$ function changes; $180^\circ/360^\circ \rightarrow$ function stays

★ THE GOLDEN ALLIED-ANGLE METHOD ★ (1) Is the reference $90^\circ/270^\circ$ (change function) or $180^\circ/360^\circ$ (keep function)? →
(2) Which quadrant does the full angle land in? Apply ASTC for the sign. Done.

🎯 WHAT THE EXAM LOVES — Marks-Grabbers

- **Allied-angle simplification** (e.g. find $\sin 150^\circ$, $\cos 240^\circ$) — guaranteed, use the golden method.
- **Prove an identity** using $\sin^2 + \cos^2 = 1$ — classic 3-4 mark question.
- **Find remaining ratios** given one (e.g. $\sin\theta = 3/5$, find rest) — very common.
- **Signs of functions in a quadrant** — 1-mark MCQ, free marks.
- **Domain / range / period** — short theory MCQ.
- **Negative-angle evaluation** — quick, just apply even/odd.

"Connect first, then improve." — Lock the angle table + ASTC + 3 identities, and 70% of this chapter is automatic. | MISMATHE · Your MHT-CET Mentor